Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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began: July 4, 2026
last edited: October 30, 2026

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8C Consequences of Generalized Eigenspace Decomposition

Exercise 8C1

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is the operator defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that T does not have a square root.

Solution. Note that $T^2(z_1, z_2, z_3) = (z_3, 0, 0)$ and $T^3 = 0$. Suppose $S \in \mathcal{L}(\mathbb{C}^3)$ is such that $S^2 = T$. Then $S^6 = (S^2)^3 = T^3 = 0$, so S is nilpotent. By 8.16, $S^3 = 0$, but $T^2 = S^4 = SS^3 = 0$, which contradicts $T^2 \neq 0$. Therefore, T does not have a square root. ∎

Exercise 8C2

Define $T \in \mathcal{L}(\mathbb{F}^5)$ by $T(x_1, x_2, x_3, x_4, x_5) = (2x_2, 3x_3, -x_4, 4x_5, 0)$.

(a) Show that T is nilpotent.

(b) Find a square root of $I + T$.

Solution.

(a) With respect to the standard basis of $\mathbb{F}^5$, the matrix $\mathcal{M}(T)$ of T on and below the diagonal is 0, so T is nilpotent by 8.18.

(b) By part (a) and 8.16, $T^5 = 0$. As in the proof of 8.39, we may construct the polynomial

$$p(z) = 1 + \frac{1}{2}z - \frac{1}{8}z^2 + \frac{1}{16}z^3 - \frac{5}{128}z^4.$$

Squaring $p(z)$ gives

$$p^2(z) = 1 + z + z^5 q(z)$$

for some $q \in \mathcal{P}(\mathbb{F})$. Let $S = p(T)$, so $S^2 = I + T + T^5 q(T) = I + T$ by 5.17(a) and $T^5 = 0$. ∎

Exercise 8C3

Suppose V is a complex vector space. Prove that every invertible operator on V has a cube root.

Solution. Let $T \in \mathcal{L}(V)$ be invertible with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ and let $n = \dim V$. By 5.7, $\lambda_k \neq 0$ for each $k = 1, \dots, m$. Let $G_k = G(\lambda_k, T)$ for each $k = 1, \dots, m$. Then the generalized eigenspace decomposition of V is

$$V = \bigoplus_{k=1}^{m} G_k.$$

By 8.22(a), we may define $N_k = (T - \lambda_k I)|_{G_k} \in \mathcal{L}(G_k)$ for each $k = 1, \dots, m$. Since N_k is nilpotent on G_k by 8.22(b), and $\dim G_k \leq n$, we have $N_k^n = 0$ by 8.16. As in the proof of 8.39 with a cubic in place of a square, set $a_1 = \frac{1}{3}$. For each $2 \leq j \leq n-1$, the coefficient of z^j in the expansion $(1 + a_1 z + \cdots + a_j z^j)^3$ is $3a_j + f_j(a_1, \dots, a_{j-1})$ for some polynomial f_j in $a_1, \dots, a_{j-1}$. Choosing a_j such that $3a_j + f_j(a_1, \dots, a_{j-1}) = 0$, we can recursively determine $a_2, \dots, a_{n-1}$. We then obtain the polynomial

$$p(z) = 1 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

such that $p^3(z) = 1 + z + z^n r(z)$ for some $r \in \mathcal{P}(\mathbb{F})$. By the fundamental theorem of algebra, the polynomial $z^3 - \lambda_k$ has a zero in $\mathbb{C}$ for each $k = 1, \dots, m$, say μ_k. Define the operators $S_k = \mu_k p(\lambda_k^{-1} N_k)$ on G_k for each k. By 5.17(a) and $N_k^n = 0$, it follows that $S_k^3 = (\mu_k p(\lambda_k^{-1} N_k))^3 = \lambda_k I + N_k = T|_{G_k}$. For each $v \in V$, by the generalized eigenspace decomposition, we may uniquely write $v = v_1 + \cdots + v_m$ with $v_k \in G_k$ for each $k = 1, \dots, m$. Define $S \in \mathcal{L}(V)$ by $Sv = S_1 v_1 + \cdots + S_m v_m$. Since $Sv_k = S_k v_k \in G_k$ for each k, we have $S^3 v_k = S_k^3 v_k = T|_{G_k} v_k = T v_k$ for each k, and thus $S^3 = T$ on all of V. ■

Exercise 8C4

Suppose V is a real vector space. Prove that the operator $-I$ on V has a square root if and only if $\dim V$ is an even number.

Solution.
$\boxed{\Rightarrow}$ Let $S \in \mathcal{L}(V)$ be such that $S^2 = -I$. Then $\operatorname{null}(S^2 + 0S + I) = \operatorname{null}(-I + I) = \operatorname{null}(0) = V$. Since $0^2 < 4$, 5.33 gives that $\dim V$ is even.
$\boxed{\Leftarrow}$ Let $\dim V = 2n$ for some positive integer n, and let V have a basis $v_1, \dots, v_{2n}$. By the linear map lemma, there exists a linear map $S \in \mathcal{L}(V)$ such that $Sv_{2k-1} = v_{2k}$ and $Sv_{2k} = -v_{2k-1}$ for each $k = 1, \dots, n$. Then $S^2 v_{2k-1} = Sv_{2k} = -v_{2k-1}$ and $S^2 v_{2k} = S(-v_{2k-1}) = -v_{2k}$ for each $k = 1, \dots, n$. Since S^2 and $-I$ agree on a basis of V, the uniqueness part of the linear map lemma implies that $S^2 = -I$. ■

Exercise 8C5

Suppose $T \in \mathcal{L}(\mathbb{C}^2)$ is the operator defined by $T(w, z) = (-w - z, 9w + 5z)$. Find a Jordan basis for T.

Solution. Let $N = T - 2I$. Then $N(w, z) = (-3w - z, 9w + 3z) = (-3w - z)(1, -3)$, so

$$N^2(w, z) = (-3w - z)N(1, -3) = 0.$$

Let $v = (0, -1)$. Then $Nv = (1, -3) \neq 0$, so by Exercise 8A2, v, Nv is a linearly independent list of length 2, and thus is a basis of $\mathbb{C}^2$ by 2.38. We have $TNv = 2Nv$ and $Tv = Nv + 2v$, so

$$\mathcal{M}(T, (Nv, v)) = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Therefore, the list $(1, -3), (0, -1)$ is a Jordan basis for T. ■

Exercise 8C6

Find a basis of $\mathcal{P}_4(\mathbb{R})$ that is a Jordan basis for the differentiation operator D on $\mathcal{P}_4(\mathbb{R})$ defined by $Dp = p'$.

Solution. Define $v = \frac{x^4}{4!}$, so $D^4 v = 1$ and $D^5 v = 0$. By Exercise 8A2, the list $v, Dv, D^2 v, D^3 v, D^4 v$ is a

linearly independent list of length 5 and thus forms a basis of $\mathcal{P}_4(\mathbb{R})$ by 2.38. Then

$$\mathcal{M}(D, (D^4v, D^3v, D^2v, Dv, v)) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

so the list $1, x, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}$ is a Jordan basis for D. ■

Exercise 8C7

Suppose $T \in \mathcal{L}(V)$ is nilpotent and $v_1, \dots, v_n$ is a Jordan basis for T. Prove that the minimal polynomial of T is z^{m+1}, where m is the length of the longest consecutive string of 1's that appears on the line directly above the diagonal in the matrix of T with respect to $v_1, \dots, v_n$.

Solution. Let $A_1, \dots, A_p$ be the blocks as in 8.44, where each A_k is d_k-by-d_k. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors of the block A_k for each $k = 1, \dots, p$. By 8.44,

$$Tv_{k,j} = \begin{cases} \lambda_k v_{k,1} & \text{if } j = 1, \\ \lambda_k v_{k,j} + v_{k,j-1} & \text{if } j > 1. \end{cases}$$

Note that $Tv_{k,1} = \lambda_k v_{k,1}$ with $v_{k,1} \neq 0$. Since T is nilpotent, 8.17(a) implies that $\lambda_k = 0$ for all k. Then $Tv_{k,1} = 0$ and $Tv_{k,j} = v_{k,j-1}$ for $j > 1$, so

$$T^{j-1}v_{k,j} = v_{k,1}, \quad T^j v_{k,j} = 0.$$

The entries above the diagonal between the blocks A_k and A_{k+1} are all 0, so the consecutive string of 1's cannot extend beyond a single block. Therefore, each block A_k contributes a consecutive string of 1's of length $d_k - 1$, so $m + 1 = \max_k d_k$. By 8.18, the minimal polynomial of T is z^r for some r. We now claim that $r = m + 1$.

$\boxed{\leq}$ Since each $v_{k,j}$ has $j \leq d_k \leq m + 1$, we have $T^{m+1}v_{k,j} = T^{m+1-j}(T^j v_{k,j}) = 0$. Therefore, $T^{m+1} = 0$, and 5.29 yields $r \leq m + 1$.

$\boxed{\geq}$ Consider some k such that $d_k = m + 1$. Then $T^m v_{k,m+1} = v_{k,1} \neq 0$, so $T^m \neq 0$. Therefore, $r \geq m + 1$ by 5.29. Combining this with the previous inequality gives $r = m + 1$. ■

Exercise 8C8

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V that is a Jordan basis for T. Describe the matrix of T^2 with respect to this basis.

Solution. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T)$ with respect to the basis $v_1, \dots, v_n$ and let λ_k be the eigenvalue corresponding to the block A_k. By 3.43 and Exercise 8B22, $\mathcal{M}(T^2) = \mathcal{M}(T)^2$ is block diagonal with blocks $A_1^2, \dots, A_p^2$. Write $A_k = \lambda_k I + N_k$, where N_k is the matrix with 1's on the superdiagonal and 0's elsewhere. Then $A_k^2 = \lambda_k^2 I + 2\lambda_k N_k + N_k^2$, and N_k^2 has 1's on the second superdiagonal and 0's elsewhere by 3.46. Therefore, each block A_k^2 has λ_k^2 on the diagonal, $2\lambda_k$ on the superdiagonal, 1's on the second superdiagonal, and 0's elsewhere. ■

Exercise 8C9

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Explain why there exist $v_1, \dots, v_n \in V$ and nonnegative integers $m_1, \dots, m_n$ such that (a) and (b) below both hold.

(a) $T^{m_1}v_1, \dots, Tv_1, v_1, \dots, T^{m_n}v_n, \dots, Tv_n, v_n$ is a basis of V.

(b) $T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0$.

Solution. By 8.45, nilpotence of T gives a Jordan basis $u_1, \dots, u_{\dim V}$ for T. Let $A_1, \dots, A_n$ be the blocks of $\mathcal{M}(T, (u_1, \dots, u_{\dim V}))$ with diagonal entries $\lambda_1, \dots, \lambda_n$. Let $u_{k,1}, \dots, u_{k,d_k}$ be the basis vectors corresponding to the d_k-by-d_k block A_k. By 5.41 and 8.17(a), $\lambda_k = 0$, so $Tu_{k,1} = 0$ and $Tu_{k,j} = u_{k,j-1}$ for $2 \le j \le d_k$ by 8.44. Let $v_k = u_{k,d_k}$ and $m_k = d_k - 1$ for $1 \le k \le n$.

(a) is satisfied because $T^j v_k = u_{k,d_k-j}$ for $0 \le j \le m_k$, so the vectors $T^{m_k}v_k, \dots, Tv_k, v_k$ are exactly $u_{k,1}, \dots, u_{k,d_k}$ in order.

(b) is satisfied because $T^{m_k+1}v_k = T^{d_k}u_{k,d_k} = 0$ for each k, as $u_{k,1} \in \operatorname{null} T$. ■

Exercise 8C10

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V that is a Jordan basis for T. Describe the matrix of T with respect to the basis $v_n, \dots, v_1$ obtained by reversing the order of the v's.

Solution. Let $A_1, \dots, A_p$ be the Jordan blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors among $v_1, \dots, v_n$ corresponding to A_k in order. By 8.44, $Tv_{k,1} = \lambda_k v_{k,1}$ and $Tv_{k,j} = v_{k,j-1} + \lambda_k v_{k,j}$ for $2 \le j \le d_k$. In the basis $v_n, \dots, v_1$, the basis vectors corresponding to A_k appear in the reverse order $v_{k,d_k}, \dots, v_{k,1}$, with the blocks appearing in the order $A_p, \dots, A_1$. Writing $w_\ell = v_{k,d_k+1-\ell}$ for $1 \le \ell \le d_k$, we have $Tw_\ell = \lambda_k w_\ell + w_{\ell+1}$ for $1 \le \ell \le d_k - 1$ and $Tw_{d_k} = \lambda_k w_{d_k}$. Therefore, $\mathcal{M}(T, (v_n, \dots, v_1))$ is block diagonal with the blocks $A_p^t, \dots, A_1^t$. Each block A_k^t has λ_k on its diagonal, 1's on its subdiagonal, and 0's elsewhere. ■

Exercise 8C11

Suppose $T \in \mathcal{L}(V)$. Explain why every vector in each Jordan basis for T is a generalized eigenvector of T.

Solution. Let $v_1, \dots, v_n$ be a Jordan basis for T. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors corresponding to A_k in order. By 8.44, $(T - \lambda_k I)v_{k,1} = 0$ and $(T - \lambda_k I)v_{k,j} = v_{k,j-1}$ for $2 \le j \le d_k$. Hence, λ_k is an eigenvalue of T and $(T - \lambda_k I)^j v_{k,j} = 0$ for $1 \le j \le d_k$. Since $v_{k,j} \ne 0$, it follows that $v_{k,j}$ is a generalized eigenvector of T corresponding to λ_k. ■

Exercise 8C12

Suppose $T \in \mathcal{L}(V)$ is diagonalizable. Show that $\mathcal{M}(T)$ is a diagonal matrix with respect to every Jordan basis for T.

Solution. Let $v_1, \dots, v_n$ be a Jordan basis for T. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors corresponding to the block A_k, and define $U_k = \operatorname{span}(v_{k,1}, \dots, v_{k,d_k})$. By 8.44, T maps the vectors $v_{k,1}, \dots, v_{k,d_k}$ into U_k, so U_k is invariant under T, and $\mathcal{M}(T|_{U_k}, (v_{k,1}, \dots, v_{k,d_k})) = A_k$. By 5.65, $T|_{U_k}$ is diagonalizable for each k. Since A_k is upper triangular with λ_k on the diagonal, 5.41 yields that λ_k is the only eigenvalue of $T|_{U_k}$. By 5.55, $U_k = E(\lambda_k, T|_{U_k})$, so $T|_{U_k} = \lambda_k I$, and hence $A_k = \lambda_k I$. Since this holds for each block A_k, it follows that $\mathcal{M}(T, (v_1, \dots, v_n))$ is diagonal. ■

Exercise 8C13

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that if $v_1, \dots, v_n$ are vectors in V and $m_1, \dots, m_n$ are nonnegative integers such that

$$T^{m_1}v_1, \dots, Tv_1, v_1, \dots, T^{m_n}v_n, \dots, Tv_n, v_n \text{ is a basis of } V$$

and

$$T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0,$$

then $T^{m_1}v_1, \dots, T^{m_n}v_n$ is a basis of $\operatorname{null} T$.

Axler Note. *This exercise shows that $n = \dim \operatorname{null} T$. Thus, the positive integer n that appears above depends only on T and not on the specific Jordan basis chosen for T.*

Solution. Since $T^{m_k+1}v_k = 0$ for each k, $T^{m_k}v_k \in \operatorname{null} T$ for each k. Observe that $T(T^j v_k) = T^{j+1}v_k$ for each k and $0 \le j \le m_k$, where $T^{m_k+1}v_k = 0$. Therefore, the list $Tv_1, \dots, T^{m_1}v_1, \dots, Tv_n, \dots, T^{m_n}v_n$ spans $\operatorname{range} T$ and is linearly independent since it is part of the given basis. Consequently, the list forms a basis of $\operatorname{range} T$, and hence $\dim \operatorname{range} T = \sum_{k=1}^{n} m_k$. Because $\dim V = \sum_{k=1}^{n}(m_k + 1)$, the fundamental theorem of linear maps gives

$$\dim \operatorname{null} T = \sum_{k=1}^{n}(m_k + 1) - \sum_{k=1}^{n} m_k = n.$$

Since $T^{m_1}v_1, \dots, T^{m_n}v_n$ is part of the given basis, it is a linearly independent list of length n in $\operatorname{null} T$, wherefore it forms a basis of $\operatorname{null} T$ by 2.38. ■

Exercise 8C14

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that there does not exist a direct sum decomposition of V into two nonzero subspaces invariant under T if and only if the minimal polynomial of T is of the form $(z - \lambda)^{\dim V}$ for some $\lambda \in \mathbb{C}$.

Solution. Let p be the minimal polynomial of T and let $n = \dim V$.

$\boxed{\Rightarrow}$ Suppose no direct sum decomposition of V into two nonzero subspaces invariant under T exists. By 8.46, there exists a Jordan basis $v_1, \dots, v_n$ of V with blocks $A_1, \dots, A_q$. Define U to be the span of the basis vectors corresponding to the first block A_1 and W to be the span of the remaining basis vectors. By 8.44, T maps the basis vectors corresponding to each block into their own span, so U and W are invariant under T. Since $U + W$ contains the basis vectors and $\dim U + \dim W = n$, it follows that $V = U \oplus W$ by 3.94. Since $U \neq \{0\}$, the hypothesis implies that $W = \{0\}$. Hence, $\mathcal{M}(T, (v_1, \dots, v_n)) = A_1$

is a single n-by-n block. Let $\lambda = \lambda_1$. By 8.44, $(T - \lambda I)v_1 = 0$ and $(T - \lambda I)v_j = v_{j-1}$ for $2 \leq j \leq n$, wherefore $(T - \lambda I)^n = 0$ and $(T - \lambda I)^{n-1}v_n = v_1 \neq 0$. By 5.29, p divides $(z - \lambda)^n$, so $p = (z - \lambda)^r$ for some positive integer $r \leq n$. On the other hand, if $r \leq n - 1$, then $(z - \lambda)^{n-1}$ is a multiple of p, so $(T - \lambda I)^{n-1} = 0$ by 5.29, contradicting $(T - \lambda I)^{n-1}v_n = v_1 \neq 0$. Hence, $r = n$, so $p = (z - \lambda)^n$.
$\boxed{\Leftarrow}$ Suppose $p = (z - \lambda)^n$ for some $\lambda \in \mathbb{C}$. Let $V = U \oplus W$ with U and W invariant under T. Since $(T-\lambda I)^n = 0$ and invariance of U under T gives invariance under $T-\lambda I$, it follows that $(T-\lambda I)^{\dim U} = 0$ on U by 8.16. A similar argument shows that $(T - \lambda I)^{\dim W} = 0$ on W. Let $M = \max(\dim U, \dim W)$. Then $(T-\lambda I)^M = 0$ on $U + W = V$, so $M \geq n$ by 5.29. By 3.94, $\dim U + \dim W = n$, so $M \geq \dim U + \dim W$, forcing $\min(\dim U, \dim W) = 0$. Hence, either $U = \{0\}$ or $W = \{0\}$, proving that no direct sum decomposition into two nonzero invariant subspaces exists. ■